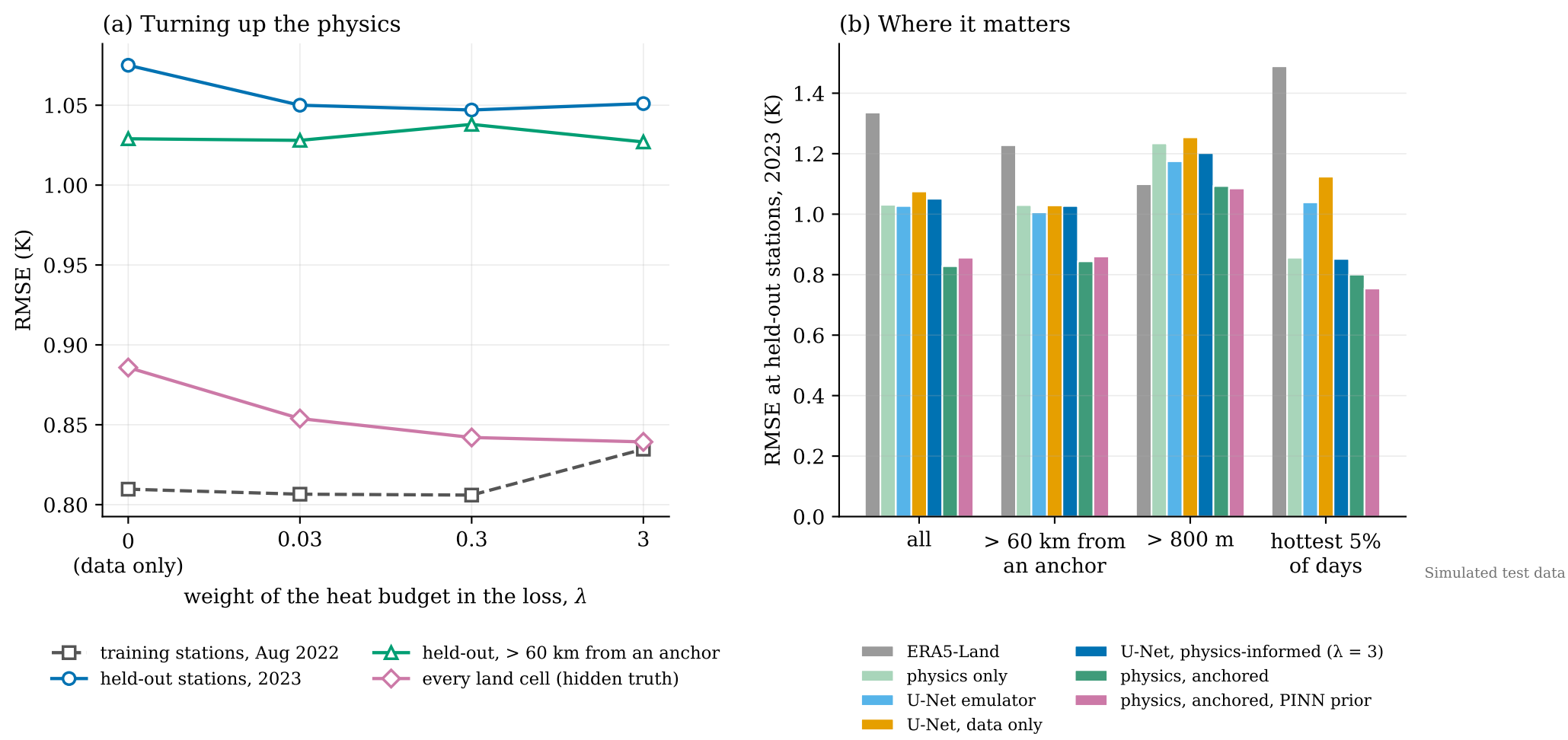
(a) Turning up the physics
RMSE (K)
1.05
1.00
0.95
0.90
0.85
0.80
0
(data only)
0.03
0.3
3
weight of the heat budget in the loss, λ
training stations, Aug 2022
held-out stations, 2023
held-out, > 60 km from an anchor
every land cell (hidden truth)
(b) Where it matters
RMSE at held-out stations, 2023 (K)
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
all
> 60 km from an anchor
> 800 m
hottest 5% of days
Simulated test data
ERA5-Land
physics only
U-Net emulator
U-Net, data only
U-Net, physics-informed (λ = 3)
physics, anchored
physics, anchored, PINN prior